

File Inventory for GraphSAGE PPI Benchmark

Investigation

Goal

Identify exactly how the GraphSAGE PPI dataset (features, labels, graph structure) was generated, to understand the label leakage that makes the benchmark trivial.

PRIMARY DATA (the thing we're investigating)

GraphSAGE PPI dataset

`/mnt/user-data/uploads/graphsage_ppi.zip` — canonical dataset from snap.stanford.edu/graphsage/ppi.zip

Contains: ppi-G.json, ppi-feats.npy, ppi-class_map.json, ppi-id_map.json, ppi-walks.txt 56,944 nodes, 818,716 edges, 50 binary features, 121 binary labels

DGL PPI dataset

`/mnt/user-data/uploads/dgl_ppi.zip` — pre-processed version from data.dgl.ai/dataset/ppi.zip Contains: {train,valid,test}_{graph.json,feats.npy,labels.npy,graph_id.npy} Same data but: z-scored features, directed edges, self-loops on all nodes, split into files **STATUS: FULLY REPRODUCED** — transformation from GraphSAGE to DGL confirmed by byte-identical

GRAPH TOPOLOGY SOURCES

OhmNet tissue networks

`/home/claude/ohm/bio-tissue-networks/` — 144 tissue-specific PPI edgelist from SNAP Binary edges with Entrez Gene IDs. Source: BioGRID physical interactions. **STATUS: All 24 GraphSAGE graphs matched to specific**

OhmNet tissues

OhmNet tissue labels

`/home/claude/ohm/bio-tissue-labels/` — 503 tissue-specific GO annotation files **STATUS: NOT the source of**

GraphSAGE labels (0/121 match)

OhmNet code

`/mnt/user-data/uploads/ohmnet-master.zip` — embedding algorithm only, no data preprocessing **STATUS: No label/feature generation code found**

GIANT/HumanBase gold standards

`/mnt/user-data/uploads/HumanBase-blood.dat` — C1-C4 classified gene pairs (4,730 genes)

`/mnt/user-data/uploads/HuamnBase-kidney.dat` — same format, same 4,730-gene universe **STATUS: Confirmed**

OhmNet 4,510 genes $\subset$ GIANT 4,730 genes. C1 co-annotation structure is consistent with labels but not independently informative.

GIANT network sample

`/mnt/user-data/uploads/blood_sample_tsv.gz` — sampled top/bottom edges from GIANT blood 25,689 genes, 34.6M weighted functional edges — completely different from OhmNet's 3,326-gene, 54K-edge binary PPI network. **STATUS: GIANT functional networks are NOT the source of OhmNet graph topology**

FEATURE PROVENANCE SOURCES

MSigDB gene set collections

`/mnt/user-data/uploads/msigdb_v5_2_files_to_download_locally.zip` — MSigDB v5.2 (Sep 2016)
`/mnt/user-data/uploads/msigdb_v5_2_chip_files_to_download_locally.zip` — chip mappings
`/mnt/user-data/uploads/msigdb_v5_1_files_to_download_locally.zip` — MSigDB v5.1
`/mnt/user-data/uploads/msigdb_v6_0_files_to_download_locally.zip` — MSigDB v6.0 **STATUS: 50/50 features**

CONFIRMED from MSigDB v5.2

Columns 0-29: C1 (positional gene sets, 30 chromosomal bands)
Columns 30-49: C3 (motif gene sets, 20 TF binding motifs)
NO C7 features despite paper claiming "immunological signatures"
Column 10: all-zero (no gene in universe is a member)

LABEL PROVENANCE SOURCES

GOA (Gene Ontology Annotation) files from EBI

`/mnt/user-data/uploads/gene_association_goa_human_155.gz` — March 2016
`/mnt/user-data/uploads/gene_association_goa_human_156.gz` — April 2016
`/mnt/user-data/uploads/gene_association_goa_human_157.gz` — May 2016
`/mnt/user-data/uploads/goa_human_gaf_158.gz` — June 2016
`/mnt/user-data/uploads/goa_human_gaf_159.gz` — July 2016 (BEST MATCH)
`/mnt/user-data/uploads/goa_human_gaf_160.gz` — September 2016 **STATUS: Best result 77/121 columns at ≥99%**

using v159

GOA support files

`/mnt/user-data/uploads/2016-06-01-gp2protein_human.gz` — UniProt-to-UniProt (not useful)
`/mnt/user-data/uploads/2016-06-01-gp2protein_geneid.gz` — Entrez-to-UniProt mapping **STATUS: Used to bridge**

GOA UniProt accessions to Entrez IDs

GO ontology structure

`/mnt/user-data/uploads/2016-06-01-go.obo` — GO hierarchy for is_a propagation
`/mnt/user-data/uploads/2016-06-01-go.owl` — OWL format (not used)
`/mnt/user-data/uploads/2016-06-01-gene_ontology.obo` — alternative format **STATUS: Used for true-path-rule propagation (is_a only; part_of makes it worse)**

NCBI gene2go

`/mnt/user-data/uploads/2016-12-23-gene2go_human_tsv.gz` — Dec 2016 Wayback Machine snapshot
`/mnt/user-data/uploads/2026-08-14-gene2go_human_tsv.gz` — current version **STATUS: Dec 2016 covers**

4,504/4,510 OhmNet genes (vs GOA's 4,481) but is 4 months too late. Current version is 10 years too late.

No mid-2016 snapshot exists.

Bioconductor org.Hs.eg.db (frozen gene2go snapshots)

`/mnt/user-data/uploads/bioconductor-annotation-org_Hs_eg_db_3_0_0_tar.gz` — Oct 2014
`/mnt/user-data/uploads/bioconductor-annotation-org_Hs_eg_db_3_1_2_tar.gz` — Apr 2015
`/mnt/user-data/uploads/bioconductor-annotation-org_Hs_eg_db_3_3_0_tar.gz` — May 2016

`/mnt/user-data/uploads/bioconductor-annotation-org_Hs_eg_db_3_4_0_tar.gz` — Oct 2016 **STATUS: All worse than direct GOA matching. Earlier dates are worse, not better.**

Greene 2015 supplementary files

`/mnt/user-data/uploads/Greene2015_Table6.xlsx` — 973 expert-curated GO BP terms

`/mnt/user-data/uploads/Greene2015_Table9.xlsx` — GO-to-tissue mappings

`/mnt/user-data/uploads/Greene2015_sup.pdf` — supplementary figures **STATUS: The 973 Greene terms filtered to ≥15 annotated genes gives 117-118, not 121. The 121 labels are NOT restricted to Greene's curated set.**

Papers

`/mnt/user-data/uploads/OhmNet.pdf`

`/mnt/user-data/uploads/Greene2015.pdf`

GENERATED ARTIFACTS

Gene identity recovery

`/mnt/user-data/outputs/node2gene_corrected.json` — node-to-Entrez mapping for all 24 graphs 4,278 genes mapped. 19 swaps corrected via MSigDB feature matching. **STATUS: 50/50 features match MSigDB v5.2 exactly after corrections**

Reports

`/mnt/user-data/outputs/ppi_lookup_table_report.md` — leakage measurement

`/mnt/user-data/outputs/ppi_gene_identity_report.md` — gene recovery methodology

`/mnt/user-data/outputs/ppi_final_report.md` — earlier comprehensive report

Scripts

`/mnt/user-data/outputs/sample_giant_network.sh` — local script to sample GIANT networks